

HARSHA KARTHIKEYAN

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

.NET Software Engineer with 4 years of experience building and supporting high-volume, cloud-based applications. Skilled in C#, .NET 8, ASP.NET Core Web API, SQL Server, Azure, and Kafka, with expertise in REST APIs, background processing, and event-driven architectures. A collaborative team player who builds strong working relationships and consistently contributes to shared goals.

TECHNICAL SKILLS

Languages: C#, SQL, TypeScript, JavaScript

Frameworks: .NET 8, ASP.NET Core Web API, Entity Framework Core, Angular, LINQ

Cloud & DevOps: Microsoft Azure (App Service, WebJobs, File Storage, Application Insights, KeyVault), Azure DevOps CI/CD

Database & Messaging: Microsoft SQL Server, Kafka

Development Practices & Tools: Agile, Scrum, Git, GitHub Copilot, Postman, Swagger, JIRA

Architecture: OOP, SOLID Principles, REST APIs, Microservices, Multi-tenancy, Dependency Injection, JWT, Caching

Testing: Unit Testing, xUnit

PROFESSIONAL EXPERIENCE

Optum | Jul 2022 - Jun 2026

Project 1: Notification Generation System (NGS)

Overview: A healthcare SaaS platform that generates letters for UnitedHealthcare health plan members and healthcare providers. It processes more than 30 million letters annually across 10+ business units, in a regulated environment with strict audit requirements.

Objective: Built to replace a third-party letter generation tool, bringing letter generation in-house.

Tech stack: .NET, Microsoft SQL Server, Angular, Apache Kafka, Microsoft Azure

Software Engineer (Mar 2024 - Jun 2026)

- Developed ASP.NET Core Web APIs and Microsoft SQL Server features to onboard two business units, processing 250K+ notifications a week for 3,000+ concurrent users.
- Built an Angular management dashboard and integrated it with ASP.NET Core Web APIs, enabling stakeholders to update configurations without code changes.
- Automated a manual letter intake process using C# .NET background services and SQL Server, processing 7,000–10,000 letters daily and saving 5–8 minutes of manual effort per letter.
- Migrated 40 Apache Kafka topics from on-premises to Confluent Cloud (hosted on GCP), configuring ACLs and certificate-based authentication with zero message loss.
- Contributed to migrating .NET applications from Windows Server VMs to Microsoft Azure (App Service, Key Vault) and consolidated 7 configuration databases into a single multi-tenant database, reducing infrastructure costs.
- Monitored applications and resolved production incidents using Azure Application Insights, SQL Server query analysis and optimisation.
- Participated in code reviews and supported production deployments.

Associate Software Engineer II (Jul 2023 - Mar 2024)

- Built ASP.NET Core REST APIs using C#, EF Core, and SQL Server with SOLID principles, integrating them with Angular front-end components.
- Upgraded 36 background services from .NET 6 to .NET 8, replacing unsupported libraries such as iTextSharp.
- Wrote comprehensive xUnit unit tests for all user stories and supported end-to-end integration testing.
- Created C# .NET background jobs to generate and process PDF letters using data from SQL server.
- Onboarded a new business unit through configuration-driven enhancements and Kafka and API integrations.

Project 2: Appeals and Grievance Tracking System

Overview: A system that manages appeals and grievances submitted by UnitedHealthcare health plan members and in-network healthcare providers, covering intake, review and decision-making.

Objective: Built to automate the appeals and grievance process, which was previously manual.

Tech stack: .NET, Angular, Microsoft SQL Server

Technology Development Program (TDP) II (Jul 2022 - Jul 2023)

- Built the review/audit page in Angular and ASP.NET Core REST APIs which is used to validate submitted appeals and grievances as part of the application workflow.
- Integrated with an external document archive API (Doc360) so users could download case attachments such as PDFs from the UI, and maintained and updated SQL Server stored procedures.
- Collaborated with cross-functional teams in an Agile environment, actively participating in all Scrum ceremonies to drive the continuous delivery of features.

EDUCATION

Bachelor of Technology - Electronics and Instrumentation Engineering, SASTRA University

2018 - 2022

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) | Generative AI Certification (AI Dojo) | UnitedHealthcare Medicaid & Retirement Certification | AHIP AHM 250 – Healthcare Management - An Introduction